



A. Robots & Automata

1. Read the text below.

A **robot** is a computer-programmed machine that can do tasks automatically and is made up of different parts. They can perform actions, manipulate objects, etc. in precise and often repetitive ways. Man-like machines, called **automata**, whose basic components are similar to a human body.

- They have **joints**, mechanical links that allow parts to move.
- The movement is created by **actuators**, their heart and muscles are the pneumatic or electric motors or systems.
- At the tip of a robotic arm, you can often find **end effectors**, tools or grippers, which help the robot pick things up or perform tasks.
- To understand the environment, robots use **sensors**, such as cameras or infrared controls, which transmit information to the central nervous system in order to locate objects or adjust movements.
- Finally, robots depend on a **computer system**, the brain, which is a program or computer that controls the robot's actions and processes information from the sensors.

2. Complete the gap fill with the bolded words from above.

Researchers at Grand University are developing a machine which they hope will recognize objects with cameras that will work as

- 1.) sensors, and retrieve objects with an arm that will be its
- 2.) end effectors. Although the arm will have
- 3.) joints that will link its muscles and an electric motor that will be the
- 4.) actuators, this new
- 5.) robot won't move like a human, i.e., it won't be like the
- 6.) automata of science-fiction films. It will be desk based, no walking or climbing stairs.
- The goal is to discover how the brain performs 'multi-tasking' and to use that information to develop the
- 7.) computer system to create a robot that can think for itself.

B. Robot Functions

1. Read the text below

Do you know where the word *robot* comes from? The Czech word, *robota*, which means compulsory labour. Robots are instrumental in activities which can be too precise, boring or dangerous for human beings.

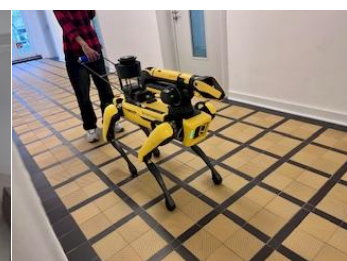
Healthcare: **Surgical robots** are programmed to assist human surgeons in very delicate microsurgery operations or to mimic the surgeons' movement in telesurgery operations.

Safety: **Mobile robots** are vehicles controlled by human operators to handle hazardous materials, defuse bombs and more.

Space: **Space probes** are unpiloted spaceships used to explore space and **planetary robots** are remotely-operated vehicles.

Industry: **Robotic arms** can be telescopic or bending arms that are widely used in the automobile industry to weld, paint or assemble car parts. They are also used in the electronic assembly of microchips when steady and precise movements are essential.

2. Match the images below to the types of robots from above.



1.)

2.)

3.)



4.)

5.)

6.)



Sprachenzentrum



C. Artificial Intelligence

1. Read the text below.

The science that tries to recreate the human thought process and build machines to perform tasks that usually require human intelligence is called **Artificial Intelligence (AI)**. It has several uses.

Neural networks: concepts in computer programming that are designed to replicate the human ability to handle ambiguity by learning from trial and error. They use silicon neurons to imitate the functions of brain cells and usually involve a great number of processors working at the same time.

Expert systems: the term given to computer software that mimics human reasoning, by using a set of rules to analyze data and reach conclusions.

Androids: anthropomorphic robots designed to look and behave like a human being. Most can walk, talk, understand human speech and 'learn' from the environment by storing information and adapting their behavior according to previous experiences.

2. Complete the gap fill with the bolded words from above.

Also known as knowledge-based systems, 1.) expert systems _____
reflect the structure of an expert's thoughts.

These are effective at predicting patterns and are often used to mimic brain activity,

2.) neural networks _____.

The term 3.) Artificial Intelligence _____ is defined as the automation of intelligent behavior, but can 4.) AI _____ really be intelligent?